

Linux Cheat Sheet

Common

- **man** : view the manual pages of a command
 - `man ls`
 - `man /usr/.../file` to read manual files using file path
- **whatis** : view a single line description of a command
 - `whatis ls`
- **info** : view detailed descriptions regarding a command
 - `info ls`
- **whereis** : find where something is located
 - `whereis ls` shows the location of `ls` command and its manual page
 - `whereis -m ls` shows the location of manual page of `ls`. It can be read directly by `man`
- **which** : shows the default location of something
 - `which python` if multiple python interpreters are installed shows the default one

Working with users

- **whoami** : shows the username
- **su (or) sudo su** : switch user
 - `su` login as root
 - `su <username>` login as <username>
- **useradd** : create new user
 - `sudo useradd -mG <usergroup> <username>`
 - `m` = create home directory `G` = add to a user group (normally `sudo`)
- **usermod** : modify user
 - `sudo usermod -aG <usergroup> <username>`
 - `aG` = add to supplementary groups
- **userdel** : remove user
 - `sudo userdel -r <username>`
 - `r` = remove home directory
- **passwd** : change password
 - `sudo passwd <username>` (login as root user if you want to ignore the password specifications)

Working with directories

- `pwd` : print working directory
- `cd` : change directories
 - `cd Desktop`
 - `cd (or) cd ~` move to root directory
 - `cd ..` move to the parent directory
 - `cd -` move to the previous directory
- `ls` : list out the items inside a certain folder
 - `ls -a` view all files including hidden ones
 - `ls -l` view a long list with all the details
 - `ls -h` view file size in human readable way. Should be used with `-l`
 - `ls -lha (or) ls -l -h -a`
- `mkdir` : make a directory
 - `mkdir -p` make parent directories if needed
 - `mkdir Desktop/project01/part01` may not work if the parent directories are not present. But `mkdir -p Desktop/project01/part01` will solve that issue
- `rmdir` : remove a directory
 - `rmdir -p` recursively removes all the parent folders too
- `pushd` : add a directory to a previous stack of directories and change into the new directory
- `popd` : remove the current directory from the stack

Working with files

- `file` : determine the file type
 - `file -s` can be used on special files
- `touch` : create empty files
 - `touch -t` can set the date and time
 - 2020/06/29 at 15:30 → `touch -t 202006291530 <filename>`
- `rm` : remove files
 - `rm -i` i = interactive show a confirmation before removing the file
 - `rm -rf` r = recursive f = force can be used to remove directories which are not empty
 - `rm -v` v = verbose narrate the process
- `cp` : copy
 - `cp file1 file1.copy` copy one file
 - `cp file1 dir1` copy file to a directory
 - `cp -r dir1 dir2` copy everything in a directory to another
 - `cp f1 f2 f3 dir1` copy multiple files into a directory
 - `cp -i file1 file2` copy with interaction to prevent overwriting files

- mv : move
 - mv file1 fil2 rename files to directories
 - mv -i move with interaction

Working with file content

- head : show few lines from a file starting from the top
 - head show first 10 lines by default
 - head -n show n number of lines (n can be any number)
 - head -cn show n number of bytes (n can be any number)
- tail : show few lines from a file starting from the bottom
 - same use cases as head
 - tail -f show real-time view of a file (ctrl+C to exit)
- echo : print something on terminal
 - echo some text print some text on terminal
 - echo some text > file overwrite some text into the file
 - echo some text >> file append some text into the file
- xxd : view content inside binary files
 - xxd filename
- cat : concatenate
 - cat filename show what's inside a file
 - cat file1 filen show all the data inside multiple files
 - cat f1 f2 > f3 create a file combining other files. Same as copying to a new file
 - cat > f1 create a file name f1 and allow the user to type into the file beginning from the next line. Press ctrl+D to exit
 - cat > f1 <<stop same as before but typing stop will exit without having to press ctrl+D. name stop can be anything. (it is called a marker)
 - tac show file starting from bottom
- more & less : view large files
 - more view the file in pages
 - less view file line by line after first page
- strings : display readable ascii strings in a file
- grep : search in files
 - grep 'searchword' filename
 - grep -i 'searchword' filename ignore the case of word
 - grep -n 'searchword' filename shows the line number
 - grep -c 'searchword' filename shows the count of the word
 - grep -v 'searchword' filename hide the word and display others

- **sort** : sort items in files
 - **sort filename** sorts the content in a file
 - **sort -r filename** inverse the sort
 - it isn't going to write into the file. just display in the terminal

Working with File Permissions

- **ls -l** : the front part shows the file permissions
 - default looks like `-rw-rw-r--` → `-` `rw-` `rw-` `r--`
 - first `-` means it is a file. `d` if it was a directory
 - second `rw-` means user permission. Here user can read and write but not execute
 - third `rw-` means group permission. Here group can read and write but not execute
 - last `r--` means other (world) permission. Here others can only read.
- **chmod** : change file permission (change mode)
 - **chmod <who><operator><permission> <filename>**
 - **who** → `u`=user `g`=group `o`=others `a`=all
 - **operator** → `+`=add `-`=remove `==`=set exactly to
 - **permission** → `r`=read `w`=write `x`=execute
 - `chmod a-r` | `chmod a+wx` | `chmod a=rwx`

Other Commands

- **gzip** : compress files
- **gunzip** : decompress files
- **tar** : archive files
 - `cvf`=compress `xvf`=extract `tvf`=list contents
 - `tar archive.tar directory`
- **wget** : download files from internet
 - `wget -c https://somedlink -P ~/Desktop`
 - `c`=continue download `P`=download directory

Control Operators

- **;** commands will run separately.
 - `command1 ; command2 ; command3`
- **&** when the line ends with ampersand the command will run in the background.
 - `sleep 20 &` (sleep command will sleep the terminal for n times but in here it only runs in the background, so the terminal will be usable)
- **\$?** exit code – holds the value of exit status of a program or command
 - `echo write something > file`
 - `echo $?` - will print 0 if the command ran without errors

- `&& ||` logical AND and OR
- `#` shell comment
- `\`
 - escape special characters
 - `echo hello\;` (will print ;)
 - to multiline commands
 - `echo hello\`
 - `this is\`
 - multiline command